

[Description](#) [Editorial](#) [Solutions](#) [Submissions](#)

121. Best Time to Buy and Sell Stock

Solved

Easy Topics

You are given an array `prices` where `prices[i]` is the price of a given stock on the i^{th} day.

You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.

Return *the maximum profit you can achieve from this transaction*. If you cannot achieve any profit, return `0`.

Example 1:

Input: `prices = [7,1,5,3,6,4]`**Output:** 5**Explanation:** Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5.

Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.

Example 2:

Input: `prices = [7,6,4,3,1]`**Output:** 0**Explanation:** In this case, no transactions are done and the max profit = 0.

</> Code

Java Auto

```
1 class Solution {
2     public int maxProfit(int[] prices) {
3         int minPrice = Integer.MAX_VALUE;
4         int maxProfit = 0;
5
6         for (int price : prices) {
7             if (price < minPrice) {
8                 minPrice = price;
9             } else {
10                 maxProfit = Math.max(maxProfit, price - minPrice);
11             }
12         }
13
14         return maxProfit;
15     }
16 }
```

Restored from local Upgrade to Cloud Saving

☒ Testcase Test Result

You must